TRAINER KEY. DO NOT DISTRIBUTE. Day 2, Session 3: How similar is similar enough? (matching)

Source: H6-A. Four-feature matching: participants' cost 12 months after is 1,029 AED lower (reduction 1,029; interval 859 to 1,200), so the interval includes values below the 1,000 AED rule. Without manager age: -1,341 (1,263 to 1,420), which does not show which estimate is closer to the truth. 4,794 participants matched to 585 different comparison businesses (one used up to 440 times). Remaining gaps: manager age -2.97 years, cost before -161 AED. **Q1 activity:** intended pairs A-2, B-1, C-3; Business 4 unused. Differences: A 900 - 1,900 = -1,000; B 1,400 - 2,300 = -900; C 700 - 1,800 = -1,100; mean -1,000 AED. The subtraction is not a causal finding: unrecorded influences could differ.

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H6-A: The city matching study summary

GreenWaste is a fictional training case. Use this study summary as your source for today's questions.

§1 | Who, what and when?

GreenWaste provides equipment subsidies, installation help and staff training. The city study records 10,000 businesses: 4,794 participant businesses and 5,206 comparison businesses that did not receive GreenWaste. Taking part followed an efficiency-score rule, not random assignment. Annual waste management cost per business, in AED, was measured before GreenWaste and 12 months after. This study is not the pilot lottery, and it is not the analysis of businesses near the score cut-off.

§2 | How were comparison businesses selected?

For each participant business, the evaluator selected the one comparison business that was most similar before GreenWaste on four recorded features: manager age, number of staff, premises area and filtration. Each feature was scaled by its standard deviation across all city businesses. The distance between two businesses is the sum of the squared differences on these scaled features. The same comparison business could be selected more than once (matching with replacement). All 4,794 participant businesses were matched to **585 different comparison businesses**, and one comparison business was used up to **440 times**. There was no maximum acceptable distance (caliper), and no participant business was excluded. Annual waste management cost before GreenWaste was not used for matching.

Table R: Annual waste management cost 12 months after GreenWaste: participant businesses minus matched comparison businesses (AED)

Comparison	Difference (AED)	95% interval (AED)	Different comparison businesses used
Matched on four recorded features	-1,029	-1,200 to -859	585
Matched without manager age	-1,341	-1,420 to -1,263	623

Table R shows the participant businesses' mean annual cost after GreenWaste minus the matched comparison businesses' mean annual cost. With four recorded features the difference is **-1,029 AED**, which is a reduction of **1,029 AED**. The 95% interval for the reduction is **859 to 1,200 AED**. Leaving out manager age changes which comparison businesses are selected. This does not show which estimate is closer to the true effect.

§3 | What is needed for a causal reading?

After we account for the recorded features, two things must hold. First, there is no important remaining difference that affects both taking part and costs. Second, there are enough comparison businesses that are similar to the participant businesses. Costs must also be measured in the same way, and the programme must not have a large effect on comparison businesses. An unrecorded influence, such as owner motivation, could break the first condition. Motivation is not a variable in the GreenWaste data, and we do not know that it explains the result.

§4 | What can and cannot be concluded?

Table B on page 2 shows how similar the groups are on the recorded features. After matching on four features, the participant businesses' mean manager age is **2.97 years** lower than the matched comparison businesses', and their mean cost before GreenWaste is **161 AED** lower. Without manager age, these gaps are **17.12 years** and **540 AED**. Finding a match for every participant does not show that the matches are close enough. This study summary does

not give the distribution of distances, a check that the groups overlap on all features together, or an analysis of how strong an unrecorded influence would need to be to change the result.

The 95% interval allows for the reuse of comparison businesses. It treats the selected matches as given, so it does not fully allow for how the matches were chosen or for bias from unrecorded influences. The 95% interval for the reduction (**859 to 1,200 AED**) includes values below 1,000 AED. So the estimate alone does not show that our rule of at least a 1,000 AED reduction in annual waste management cost per business per year is met. The full programme costs and the effects for the whole country are not given.

Source: GreenWaste city data (10,000 businesses, measured before and 12 months after). All values are calculated from the exact records, and the figures shown are rounded. Tables R and B and paragraphs §1 to §4 are the material to inspect.

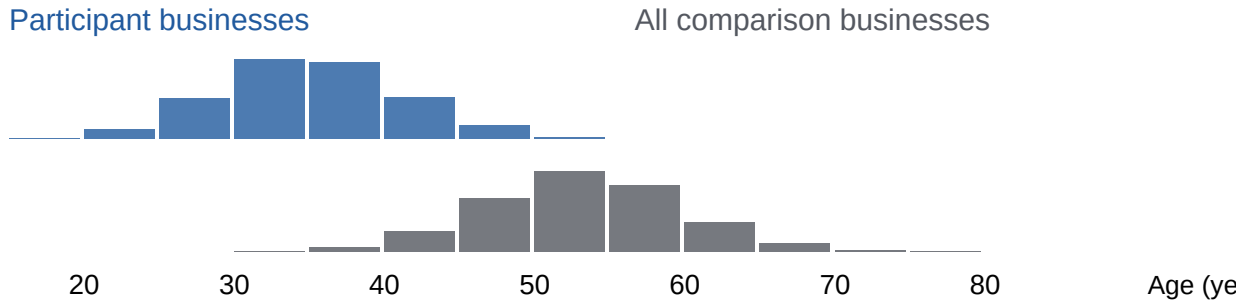
Page 2 | H6-A: How similar the matched groups are, and what the figures show

Table B: Average characteristics before GreenWaste: participant businesses, all comparison businesses and matched comparison businesses

Feature	Participant businesses	All comparison businesses	Matched comparison businesses	Gap in city standard deviations
Manager age (years)	35.77	53.90	38.74	-0.27
Staff (people)	5.32	6.66	5.44	-0.04
Premises area (100 m²)	1.81	2.34	1.82	-0.01
Filtration (share of businesses)	0.33	0.52	0.33	0.00
Annual cost before GreenWaste (AED; not used for matching)	1,431.91	2,257.58	1,592.80	-0.23

The matched comparison means count a business each time it is used. The gap is the participant mean minus the matched comparison mean, divided by the standard deviation across all city businesses. It is not a standard error and it is not a threshold for judging whether a match is good. Filtration is the share of businesses with a filter. Annual cost before GreenWaste was not used for matching.

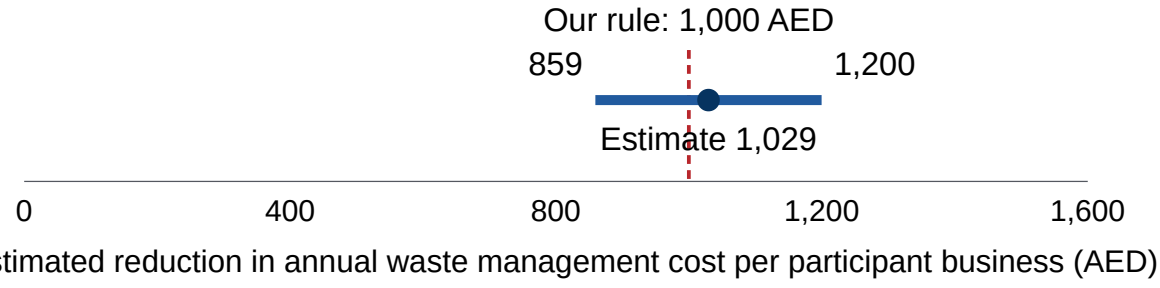
Figure 1: Manager age in participant businesses and in all comparison businesses



The bars show the share of businesses in each age group, so the two groups can be compared even though they have different numbers. Overlap in age alone does not show that the groups are similar on all four features together.

Reuse of comparison businesses and our rule

4,794 pairs does not mean 4,794 different comparison businesses. The four-feature comparison has 9,588 matched rows but only 5,379 different businesses, so some comparison businesses contribute many times.



The interval shows uncertainty about the mean matched difference, if the model's assumptions hold. It does not show the savings of individual businesses, and it does not allow for all sources of bias.

Page 3 | Q1 and Q2: work on these pages as we teach each idea

Q1 | Match three participant businesses to comparison businesses

The seven businesses in this activity are invented. They are not from the city study. Use the profile cards.

a. Match A, B and C to comparison businesses using manager age and size. Use each comparison business at most once.

	Participant business	Comparison business chosen	(b) Why did you choose this pair?
A		**2 (age 46, small)**	**Closest manager age (45 vs 46) and the same size**
B		**1 (age 39, large)**	**Closest age (38 vs 39) and the same size (large)**
C		**3 (age 51, small)**	**Closest age (52 vs 51) and the same size (small)**

c. Which comparison business is not used? **4 (age 29, large; no similar participant)**

d. Write one thing you do not know that could matter.

> Examples: owner motivation, how well the business is run, location, the type of waste, or why the business chose to take part. Anything not on the card that affects both taking part and costs.

Keep the reveal slips closed until you have recorded your choices.

You will now receive the reveal slips. A reveal slip shows a business's annual waste management cost after GreenWaste. Keep your pairs. For each pair, calculate the participant's cost minus the comparison business's cost.

Pair	Participant cost (AED)	Comparison cost (AED)	Difference (AED)
A and **2**	**900**	**1,900**	**-1,000**
B and **1**	**1,400**	**2,300**	**-900**
C and **3**	**700**	**1,800**	**-1,100**

Calculate the mean of the three differences: **-1,000** AED. Does this show that GreenWaste caused the difference? Explain why or why not.

> No. The mean difference is -1,000 AED for these invented pairs, but the pairs were matched only on age and size. Differences in unrecorded influences (e.g. motivation) could explain some or all of the gap, so subtraction is not causal identification.

Q2 | Read the result for the city study (H6-A §2 and Table R)

a. Explain the sign, the comparison, the period and the units of the four-feature result.

> Negative sign: participants' mean annual waste management cost 12 months after GreenWaste is lower than that of the selected comparison businesses by about 1,029 AED (a reduction of 1,029). Units: AED per business per year. Period: 12 months after the programme, a comparison of levels after, not a change over time.

b. Which four features were used to select the comparison businesses? Was annual cost before GreenWaste one of them?

> Manager age, number of staff, premises area and filtration. No: annual cost before GreenWaste was not used for matching.

c. Why can 4,794 participant businesses be matched to only 585 different comparison businesses?

> Each participant is matched to its nearest comparison business, with replacement and no caliper, so the same comparison business can be chosen many times (one up to 440 times); 4,794 pairs use only 585 different businesses.

d. What does this reuse mean for how we interpret the sample and the interval?

> The comparison mean counts reused businesses repeatedly, so fewer distinct comparison businesses (585) carry the comparison than the number of pairs suggests. The interval allows for the reuse but treats the matches as given: it does not fully allow for how the matches were chosen or for unrecorded bias.

Page 4 | Q3 to Q6: judge the comparison

Q3 | How similar are the matched groups? (Table B on page 2)

a. Write one measure on which the matched groups are more similar than all comparison businesses.

> Examples: filtration (0.33 vs 0.33 matched; all comparison 0.52), premises area (1.82 vs 1.81), staff (5.44 vs 5.32) or manager age (38.74 vs 35.77, against 53.90 for all comparison businesses).

b. Write one difference that remains.

> Manager age is still 2.97 years lower among participants (-0.27 SD), and annual cost before GreenWaste is 161 AED lower (1,431.91 vs 1,592.80; not used for matching).

c. Which extra check would help you judge whether the matches are close enough?

> Any one: the distribution of match distances; a joint overlap check across all four features; regions with few similar comparison businesses; sensitivity to a maximum acceptable distance (caliper); sensitivity to other matching choices.

Q4 | Leaving out age and unrecorded influences (Table R, §3 and §4)

a. Why could leaving out manager age change the estimate?

> Different feature lists select different comparison businesses. Manager age differs by 17.12 years and cost before by 540 AED between groups without it, and age is related to costs. The -1,341 vs -1,029 difference does not show which is closer to the true effect.

b. Why does making the recorded features similar not show that unrecorded influences are absent?

> Matching balances only recorded features. An unrecorded influence such as owner motivation could affect both taking part and costs; similar means on measured variables cannot show it is absent.

Q5 | Read the source yourself, then check our rule

a. Write the main finding and the paragraph or table that shows it.

> About 1,029 AED lower mean annual cost 12 months after GreenWaste among participants than among selected comparison businesses (95% interval for reduction 859 to 1,200): §2 and Table R.

b. Write the key assumption for a causal reading and the paragraph that states it.

> There is no important remaining difference (e.g. unrecorded motivation) that affects both taking part and costs: §3, first condition. (Also enough similar comparison businesses, consistent measurement and no big effect on comparison businesses.)

c. A match was found for every participant. What does that show, and what does it not show?

> It shows coverage under this method: every participant had a nearest comparison business. It does not show the matches are close enough (no distance distribution, no caliper, no joint overlap check; §4).

d. Does the whole interval meet the 1,000 AED reduction rule? Distinguish estimation uncertainty from whether a causal reading is credible.

> No: the interval 859 to 1,200 includes values below 1,000, so the estimate alone does not show the rule is met. Estimation uncertainty (the interval, allowing for reuse) is separate from whether a causal reading is credible, which depends on unrecorded influences and match quality and is not covered by the interval.

Q6 | Close the source and give your first judgement

a. Write one sentence interpreting the result, one strength of the comparison and one concern that matters for the decision.

> Example: "Matched participants' costs are about 1,029 AED lower, but the whole interval does not clear 1,000. Strength: every participant matched on four recorded features; concern: unrecorded influences like motivation, and no check of match quality." (Table R and §2-§4)

b. Choose Act, Act with conditions or Ask first, and give your reason.

> Accept Act with conditions or Ask first with a reason (e.g. "Ask first: the interval straddles 1,000 and match quality is unchecked"). Act with no conditions is hard to justify from this source.

c. Request one piece of evidence about the matching and say how the answer could change your decision.

> Example: the distribution of match distances. If many pairs are far apart, the estimate is less credible and my advice would move from Act with conditions to Ask first.

Lottery: random groups. Difference-in-differences: changes over time. Regression discontinuity: either side of a cut-off. Matching: similar businesses. Do not average these different comparisons.

Page 5 | AI exercise: compare, then verify

Stage 1: Mark the response

Naive prompt (a prompt with no context, as a busy person might type it): “Compare these two matching evaluations and advise the commissioner.” Source: H6-A (the matching study summary) and Table B. This response was written for the exercise. It is not a real AI answer.

> The four-feature evaluator compared participant businesses with comparison businesses that were similar in manager age, staff, premises area and filtration before GreenWaste. Every participant business received a match, and comparison businesses could be reused. Annual waste management cost 12 months after GreenWaste was about 1,029 AED lower, with a 95% interval for the reduction of 859 to 1,200 AED.
>
> The evaluator who left out manager age found a larger reduction of 1,341 AED. ~~A commissioner should prefer the fuller model: the extra feature removes selection bias, and finding a match for every participant business shows that the comparisons are sufficiently close.~~
>
> Both analyses point towards lower costs. The four-feature estimate is above our rule of at least a 1,000 AED reduction, ~~so the rule is securely met.~~ Verify programme costs before deciding whether to expand.

Keep the claims that H6-A supports. Cross out the claims it does not support. Write one passage from the source that changes the advice.

> Supported: the matching on four features; every participant matched; reuse; about 1,029 AED lower (859 to 1,200); Table B gaps; "verify programme costs". Crossed out: "removes selection bias", "shows the comparisons are sufficiently close", "the rule is securely met", "prefer the fuller model". Passage that changes advice: §4, "The 95% interval ... includes values below 1,000 AED" or "does not give the distribution of distances".

Stage 2: Run the improved prompt in Microsoft Copilot

Improved prompt

I commission a GreenWaste evaluation. Using only H6-A below, compare four-feature matching with the analysis that leaves out manager age. State the outcome, the comparison, the estimate and the interval. Tell me what was left off the matching list, whether costs were similar before GreenWaste, and how close the matches are. Separate the reuse of comparison businesses, the measured similarity and the unrecorded influences. Test our rule of at least a 1,000 AED reduction against the interval, not only the estimate. Ask me questions about missing information before you recommend action. Do not invent diagnostics, and do not treat matching as a lottery. Cite the paragraph or table for each fact. Here is H6-A: [paste page 1 of this worksheet and Table B from page 2].

Use only the information in H6-A, and keep your AI good-practice card beside you. If you are not using Copilot, read the prepared response on page 6 after you finish Stage 1.

a. Record one correction you verified.

> Accept any verified correction, e.g. the rule is not securely met (interval includes values below 1,000); matching every participant does not show closeness; the larger no-age estimate is not automatically less biased.

b. Record one useful question Copilot asked about missing evidence.

> Accept any specific question, e.g. "Can you supply the distribution of match distances?" or "What influences on taking part are missing from the records?"

Page 6 | Prepared response

Use this response if you are not using Copilot. Read it after you have marked the response on page 5. It was written for the exercise. It is not a real AI answer. An AI tool may answer differently.

A comparison grounded in the source

H6-A §2 and Table R compare annual waste management cost 12 months after GreenWaste. They do not compare changes over time. The four-feature matched reduction is **1,029 AED**, with an interval of **859 to 1,200 AED**. The result without manager age is **1,341 AED**, with an interval of **1,263 to 1,420 AED**. Different feature lists select different comparison businesses, and the larger result is not automatically more credible.

Every participant business received a most similar comparison business, but 4,794 participant businesses were matched to only 585 different comparison businesses (§2). That shows coverage under this method. It does not show that the matches are close enough. Table B shows a remaining age difference and a difference in cost before GreenWaste, even with four features. Annual cost before GreenWaste was not on the matching list. Similar measured means cannot remove all selection bias (§3).

The four-feature interval includes values below 1,000 AED (§4), so the estimate alone does not show that our rule of at least a 1,000 AED reduction is met. The interval does not allow for every threat to a causal reading. The full programme costs and the effects for the whole country are not provided.

Questions before recommending action

- Can you supply the distribution of match distances and evidence about overlap on all features together, or about regions with few comparison businesses?
- Why was annual cost before GreenWaste not included, and how would other defensible matching choices affect the results?
- What is known about influences on taking part that are missing from the records? Is there an analysis of how strong such an influence would need to be?
- Which population should the decision cover, what are the full programme costs, and how much uncertainty can the decision tolerate?

A conditional recommendation would separate the recorded difference, the unverified causal assumption and the uncertainty about our rule of at least a 1,000 AED reduction. It should not promise to remove bias or to expand the programme automatically.

Verify the alternative yourself

a. Check one factual claim against a named paragraph or table.

> Example: "Four-feature reduction 1,029, interval 859 to 1,200, checked against §2 and Table R." Accept any claim checked against a named paragraph/table.

b. Write one question that could change the decision, and why.

> Example: "Can you show the distance distribution and overlap? If many matches are poor, the estimate may be biased and our decision about expansion could change." Accept any specific question with a reason.